

AsBeez Loyalty Reward System (ALRS) — Product Bible

(Comprehensive Reference for Product, Engineering, Operations, Compliance, and Investors)

> **Positioning (locked):** ALRS is a **closed-loop retail loyalty program** funded by real product sales.

> **No recruiting. No sign-up fees. No subscriptions. No cash payouts.**

> Rewards are issued as **store credit** and depend on overall AsBeez sales activity.

1) Executive Summary

ALRS turns everyday shopping at AsBeez (Grocery, Bakery, Fastfood, and future AsBeez ventures) into a repeatable rewards experience. Customers earn points from qualifying purchases, which:

- 1) Accumulate as **QRP** (Qualifying Reward Points) from the customer's own purchases,
- 2) Mint **ARC** (AsBeez Reward Cells) at defined thresholds,
- 3) Enable eligibility for **GRP** (Global Reward Points) generated by **overall purchases across the AsBeez network**, and
- 4) Allow GRP to be redeemed as **store credit** under controlled rules and approvals.

ALRS is designed to be:

- **Customer-simple** (shop → earn → redeem)
- **Audit-grade** (append-only ledgers, reversals not edits, full logging)
- **Franchise-ready** (consistent rules, POS-native, centralized governance)
- **Compliance-aware** (avoid investment/income framing; store credit only)

2) Goals and Non-Goals

2.1 Product Goals

- Increase customer retention and repeat purchases (loyalty loop).
- Create network effects: rewards expand with global AsBeez sales activity.
- Keep the program financially safe via store-credit controls and reserves.
- Support franchising: consistent rules and measurable outcomes across locations.

- Provide transparency to customers: balances, history, and redemptions.

2.2 Engineering / Control Goals

- Append-only ledgers: no silent edits.
- Deterministic calculations: re-playable, idempotent processing.
- Strong governance: period locks, approvals, audit logs.
- Separation of concerns: POS vs ALRS vs Accounting vs Inventory.

2.3 Non-Goals (explicit)

- Cash payouts, external wallets, crypto rewards.
- Recruiting, referral commissions, or membership fees.
- Exposing internal margin/cost data to customers or general staff.
- Promising guaranteed rewards.

3) Terminology (Customer + Internal)

3.1 Customer-Facing Terms (approved)

- **QRP (Qualifying Reward Points):** Points earned from *your own purchases*.
- **ARC (AsBeez Reward Cell):** Earned “reward slot” created when you reach a QRP threshold.
- **GRP (Global Reward Points):** Points generated from purchases across the *entire AsBeez network*.
- **Store Credit:** Redeemable value derived from GRP (subject to rules and approval).

3.2 Internal-Only Concepts (restricted)

- Margin snapshots and reward budgets per item/transaction.
- Distribution allocation logic and reserves.
- Placement algorithm details and anti-fraud heuristics.

> **Policy:** Internal margin/cost data is **need-to-know only** and is never displayed in customer apps or cashier screens.

4) Customer Experience (Simple Explanation)

4.1 One-paragraph explanation

When you shop at AsBeez, you earn **QRP** from qualifying purchases. Once you collect **100 QRP**, you earn **1 ARC**. With at least one ARC, you may receive **GRP** generated by purchases made across the entire AsBeez network. GRP can be redeemed as **store credit** at AsBeez, subject to system rules and approval.

4.2 30-second script (staff-safe)

“Every time you shop at AsBeez, you earn points. Those points can earn reward cells, and reward cells may receive store credit from total AsBeez purchases worldwide. There’s no sign-up fee and no recruiting—just shopping and rewards.”

5) Core Mechanics (Program Logic)

5.1 QRP Earning

- QRP is earned **only from qualifying purchases** at AsBeez.
- QRP accrues to the purchasing customer/member.
- QRP is computed internally from eligible item rules (see §6).

Important rule: If a purchase is made using **store credit**, **no QRP** is awarded for that transaction.

5.2 ARC Minting

- **Threshold:** 100 QRP → mint 1 ARC.
- Customers can mint **unlimited ARCs** over time.
- ARC minting is atomic and deterministic:
- QRP consumption is recorded,
- ARC is created and assigned,
- Downstream distributions are computed.

5.3 GRP (Global Reward Points) Eligibility

- Customers become eligible for GRP once they own **at least one ARC**.
- GRP is generated by **global AsBeez purchasing activity**, not only the customer’s own purchases.
- GRP is credited under program rules (including activity/eligibility checks where applicable).

5.4 Store Credit Redemption

- GRP converts to store credit under defined rules:
- status lifecycle (pending → approved → available),
- redemption limits / frequency controls,
- fraud checks and compliance controls,
- full audit trails.

6) Reward Funding and Allocation (Internal Model)

> This section describes the intent without exposing sensitive margin details to customers.

6.1 Funding Source

- A portion of AsBeez retail profit is allocated as the ****ALRS reward pool****.
- This allocation is determined by product and transaction rules.

6.2 Allocation Buckets (high level)

The reward pool can be split conceptually into:

- ****Customer qualification bucket:**** used to award QRP (toward ARC creation).
- ****Global distribution bucket:**** used to award GRP to eligible ARC holders.
- ****Portal reserve:**** captures unused allocations (e.g., insufficient eligible levels, skipped inactive, or program controls).

> Reserve exists to keep the program financially safe, predictable, and auditable.

6.3 No Margin Disclosure

- The system stores an internal “reward budget snapshot” per transaction.
- Customer views only show points earned, not cost/profit.

7) ARC Placement (Matrix Concept)

7.1 Placement Concept (customer-friendly)

Each ARC is placed into a structured rewards network. As the network grows, ARC holders can receive GRP generated from global AsBeez purchases.

7.2 Internal Placement Rules (engineering)

- Forced ****3-wide**** structure, infinite depth conceptually.
- Placement is ****BFS (breadth-first)**** and must be:
 - efficient,
 - concurrency-safe,
 - deterministic,
 - idempotent.

7.3 Upline Distribution Concept

- When a new ARC is created, upstream eligible ARCs may receive GRP.
- If some uplines are inactive, the reward can skip to the next eligible active ARC.
- If there are not enough eligible levels, leftover allocation goes to ****portal reserve****.

8) Status & Eligibility Rules

8.1 Member Status

- Members can be active/inactive based on business rules (e.g., account standing, compliance flags, program rules).
- Inactive accounts:
 - do not receive GRP distributions,
 - may be skipped in upline selection,
 - keep historical data intact (append-only).

8.2 ARC Status

- ARC inherits eligibility behavior from owner status (with potential ARC-specific controls if needed later).

9) Data Model (Conceptual)

> Exact schema is implementation-specific; this is the conceptual contract.

9.1 Entities

- **Member**
- **Transaction / Order**
- **QRP Ledger Entry**
- **ARC**
- **ARC Placement / Node Index**
- **GRP Ledger Entry**
- **Wallet / Store Credit Account**
- **Redemption Request**
- **Approval Records**
- **Audit Logs**
- **Reserve Ledger**

9.2 Ledgers (Append-only)

- QRP Ledger: customer qualification points (earned/consumed).
- GRP Ledger: global reward points (credited/debited).
- Store Credit Ledger: redemption and adjustments (controlled, auditable).
- Reserve Ledger: leftover allocations and system captures.

9.3 Immutability Rules

- No delete/update of ledger rows.
- Corrections are reversal entries.
- Every ledger entry references a source event.

10) Event Flow (POS → ALRS)

10.1 High-Level Flow

- 1) POS completes sale
- 2) Sale triggers ALRS “earn QRP” event (if eligible payment type)

- 3) QRP ledger is credited
- 4) If QRP reaches threshold → ARC mint event
- 5) ARC placed in matrix
- 6) GRP distribution computed and posted to eligible wallets
- 7) Reserve captures leftovers
- 8) Customer sees updated balances

10.2 Key Rule: Store Credit Purchases

- If store credit used, the transaction emits a flag and ALRS suppresses QRP award.

11) Customer UI Copy (Approved)

11.1 Rewards Overview

- ****QRP Balance:**** "Points earned from your purchases"
- ****ARC Owned:**** "Reward cells you've earned"
- ****GRP Balance:**** "Global rewards available as store credit"

11.2 ARC Detail Footer

"GRP comes from global AsBeez purchases."

11.3 Redemption Disclaimer

"Store credit subject to approval."

12) POS Microcopy (Cashier-safe)

- Checkout: "Earn rewards with your purchase."
- Payment success: "You earned +X QRP from this purchase."
- ARC earned: "You earned 1 ARC. Your ARC may now receive GRP from worldwide AsBeez purchases."
- Store credit rule: "Purchases made using store credit do not earn reward points."

- Receipt footer: "AsBeez Rewards: +X QRP earned. Rewards come from global AsBeez purchases."

13) API Contract (High-Level)

> Exact endpoints can evolve; these are the functional contracts.

13.1 QRP

- Earn QRP (from completed sale event)
- Get QRP balance and next ARC threshold

13.2 ARC

- Mint ARC (threshold-based, atomic)
- List ARCs and status

13.3 GRP / Wallet

- Get GRP balances (available/pending/lifetime)
- Request redemption (store credit)
- View transaction history

13.4 Audit

- Privileged inspection endpoints restricted to admin roles

14) Governance, Controls, and Audit

14.1 Non-negotiables

- Append-only ledgers everywhere.
- Idempotency prevents double-credits.
- Period locks block backdating.
- Manager/admin approvals for sensitive actions.
- Full audit trail for approvals, overrides, unlocks, redemptions, and adjustments.

14.2 Store Credit Approval Lifecycle

- Pending → Approved → Available → Redeemed
- Each state transition is permission-gated, reasoned, and audited.

15) Compliance & Regulatory Framing

- Loyalty program, not investment.
- No fees, no recruiting, no cash payouts.
- Rewards funded by real retail sales.
- GRP depends on overall sales activity, not guaranteed.

16) Security & IP Protection (ALRS-specific)

- Placement algorithm internals and distribution logic are crown-jewel IP.
- Repo segmentation restricts ALRS engine access.
- Least privilege for developers and staff.
- AI policy: do not paste crown-jewel logic externally.

17) Testing Strategy

- Unit tests: eligibility, thresholds, distributions, inactive skipping, reserve capture.
- Integration tests: sale → QRP → ARC → GRP end-to-end.
- Load tests: BFS placement and idempotency under concurrency.

18) Analytics & KPIs

Customer KPIs

- Active members, repeat purchase rate, QRP earn rate, ARC mint rate, GRP redemption rate.

Program KPIs

- Reserve flows, store credit exposure, coverage of eligible uplines, inactive skip frequency, fraud flags per 1,000 orders.

19) Rollout Plan (Recommended)

- Pilot one branch; monitor correctness, support volume, redemption patterns.
- Expand gradually across branches.
- Franchise rollout only after stable audit and closing controls.

20) Customer FAQs (Short)

- ****Do I pay to join?**** No.
- ****Do I need to recruit?**** No.
- ****Is GRP guaranteed?**** No; depends on overall sales activity.
- ****Is it cash?**** No; store credit only.
- ****Do store credit purchases earn QRP?**** No.

21) Approved Phrases vs Prohibited Phrases

Approved

- "Shop and earn rewards."
- "Rewards come from global AsBeez purchases."
- "Store credit subject to approval."
- "No fees. No subscriptions. No recruiting."

Prohibited

- "Guaranteed earnings"
- "Passive income"
- "Investment return"
- "Profit share"
- "Commission"

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